GreenLCC India: what it is and what it needs

The problem, how the tool works, every input file it reads, and what it gives back · 23 Oct 2026

In one paragraph

GreenLCC is a decision tool for Indian buildings that want an **IGBC green rating**. Most projects pick green measures (solar, efficient air-conditioning, better glazing, LED lighting, water fixtures) by **lowest upfront price**. GreenLCC instead costs every measure over the building's **whole life**: purchase, maintenance, parts that must be replaced, energy and water saved as tariffs rise, and leftover value at the end. It then tests every legal combination of measures to find the one that reaches the target rating for the **least lifetime cost**, and checks how safe that answer is if electricity prices and interest rates change.

The problem it solves

- **First-cost bias.** Green buildings cost a few percent more upfront, and tenders reward the lowest price, so options that save far more over 25 years lose to cheaper ones.
- **Rising, uncertain tariffs.** Commercial electricity tariffs keep changing, while credit for exported rooftop solar is low (about Rs 2.82/kWh in Maharashtra). A single payback number hides that risk.
- **Points chased blindly.** Ratings bring marketing value and, in some states, extra buildable floor area, but the cheapest points to buy are rarely the cheapest over the building's life.
- **Manual, unequal comparisons.** Equipment lasts different lengths of time, and vendor quotes arrive as PDFs that people retype into spreadsheets.

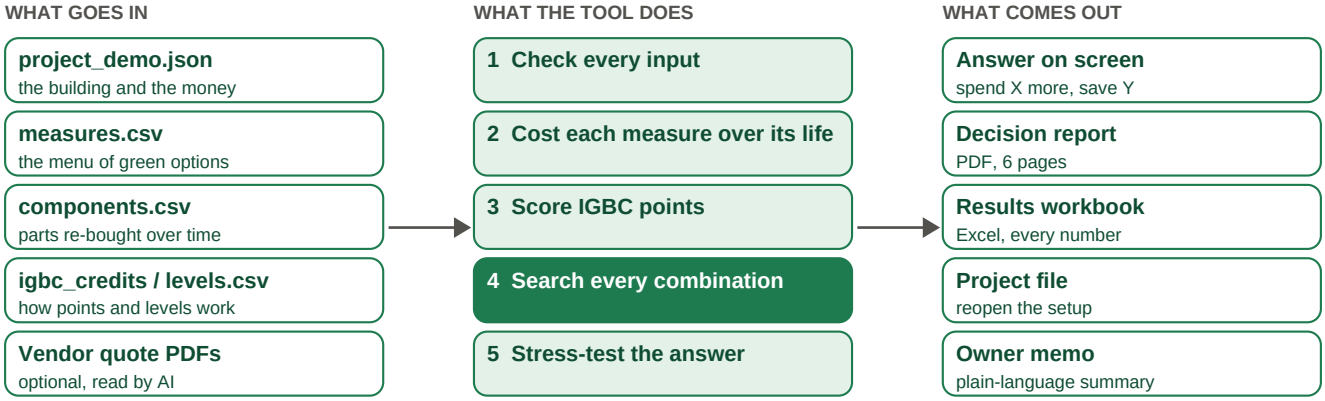
Who it is for

Building owners and developers, architects and MEP consultants justifying a specification, green-rating consultants, and procurement teams comparing bids on lifetime cost rather than price.

What it found for the demo building

Question	Answer (illustrative Pune office, Gold target, 25 years)
Which measures?	VRF air conditioning, Low-e double-glazed unit, LED lighting + daylight sensors, Energy meters + BMS, Cool roof coating, Rooftop solar 150 kWp, Low-flow water fixtures
Rating reached	Gold, 74 points
Versus cheapest-upfront route	Rs 1.10 Cr more upfront, Rs 82.3 L better over 25 years
How safe?	Wins in 97% of 3,000 simulated futures
Carbon	About 7,408 tonnes of CO_2 avoided

How it works



Step	What happens, in plain words
1 Check every input	Each file is checked row by row (types, ranges, links between files). All problems are listed at once in plain words; nothing is silently accepted.
2 Cost each measure	For every year of the study period: upfront cost in year 0, maintenance growing each year, parts re-bought when they wear out, minus energy and water savings (tariffs rising, equipment slowly degrading), minus leftover value at the end. Each year is converted to today's money and added up: that total is the **lifetime cost**.
3 Score IGBC points	Fixed credits add their points. Energy, renewable and water credits are scored from the combined savings of the whole package against threshold tables, so nothing is counted twice. Mandatory credits are checked.
4 Search every combination	One option per group (one HVAC system, one solar size...) or none; other measures on or off. For the demo that is 1,152 combinations, all checked, so the answer is exact. Solar must fit the roof.
5 Stress-test the answer	Tipping points (where the ranking flips), one-at-a-time sensitivity, and thousands of simulated futures with random tariff growth, discount rate, equipment prices and solar output.

The core formula

Lifetime cost (LCC) = sum over years t = 0 ... N of (costs in year t - savings in year t) / $(1 + d)^t$, where d is the discount rate. Cost per year (EAC) = LCC x d / (1 - $(1 + d)^{-N}$). All money is in nominal rupees; prices are year-1 prices that rise from year 2.

AI features

Two optional helpers use Claude when an API key is set: reading a vendor quote PDF into draft fields (always reviewed by a person), and polishing the owner memo (it only sees computed numbers and may not change them). Without a key, a built-in reader and a template are used.

The input files

All inputs live in the **data/** folder as plain CSV and JSON, so they can be edited in Excel or any text editor. Everything in the demo is **illustrative**: realistic in size, but not verified prices, tariffs or official IGBC tables.

File	What it describes	Format	Demo contents
project_demo.json	One building: areas, energy and water use, tariffs, finance, IGBC target, FAR bonus	JSON, one object	Pune Tech Office (demo)
measures.csv	The menu of green measures and what each costs and saves	CSV, one row per measure	12 measures
components.csv	Parts that wear out and must be re-bought	CSV, one row per part	12 parts
igbc_credits.csv	How each IGBC credit earns points	CSV, one row per credit	8 credits
igbc_levels.csv	Points needed for each rating level	CSV, one row per level	Certified 50, Silver 60, Gold 70, Platinum 80
sample_quotes/*.pdf	Vendor quotations for the AI importer (optional)	PDF	3 sample quotes
Project file (.json)	Everything above in one file, saved from the Report page	JSON	made by the tool

project_demo.json: the building and the money

Field	Meaning	Unit	Demo value	Rule
name	Project name	text	Pune Tech Office (demo)	shown on screens and reports
city	City	text	Pune	
building_type	Building type	text	Commercial office	
plot_area_m2	Plot area	m2	5000	> 0
fsi	FSI / FAR allowed on the plot	ratio	2.4	> 0
bua_m2	Built-up area	m2	12000	> 0
roof_area_m2	Usable roof area (limits solar size)	m2	2500	>= 0
baseline_energy_kwh	Electricity use of the code-minimum building	kWh / yr	1.8e+06	> 0
baseline_water_kl	Water use of the code-minimum building	kL / yr	14000	>= 0
study_period	Years the costs are counted over	years	25	5 to 60
discount_rate	Discount rate (nominal)	fraction	0.08	0 to 0.30
inflation	General inflation (replacement prices)	fraction	0.05	0 to 0.20
grid_tariff	Electricity tariff in year 1	Rs / kWh	11	> 0
tariff_escalation	Yearly tariff growth	fraction	0.05	-0.05 to 0.20
tariff_escalation_sd	Uncertainty of that growth (1 sd)	fraction	0.02	0 to 0.10
export_tariff	Credit for solar exported to the grid	Rs / kWh	2.82	>= 0
export_escalation	Yearly growth of the export credit	fraction	0	optional
water_tariff	Water + sewage tariff	Rs / kL	50	>= 0
water_escalation	Yearly water tariff growth	fraction	0.06	optional
pv_yield_kwh_per_kwp	Solar output per kWp per year	kWh / kWp	1450	> 0
pv_self_consumption	Share of solar used on site (rest exported)	0 to 1	0.9	

Field	Meaning	Unit	Demo value	Rule
pv_area_per_kwp_m2	Roof area needed per kWp	m2 / kWp	10	optional
grid_emission_factor	Grid CO_2 per MWh	tCO_2 / MWh	0.71	>= 0
target_level	Default IGBC target	level name	Gold	must exist in igbc_levels.csv
base_points	Points the base design already earns	points	52	>= 0
far_bonus_pct	Extra FAR per level	% by level	Silver 3%, Gold 5%, Platinum 7%	levels must exist
far_net_value_per_m2	Net value of each extra m2 built	Rs / m2	30000	>= 0
use_far_benefit	Count the FAR value?	true / false	True	

measures.csv: the menu of green options

One row per option. Costs are **extra over a code-minimum baseline** (doing nothing extra), so a VRF row holds the VRF price minus the split ACs it replaces.

Column	Meaning	Rule
measure_id	Unique short code, e.g. PV_100	required, unique
name	Readable name shown on screens	required
exclusive_group	Measures in the same group are alternatives (only one chosen). Blank = independent	optional
kind	efficiency, renewable, water or none: decides which savings credit it feeds	required
credit_id	IGBC credit it earns (must exist in igbc_credits.csv)	optional
points_fixed	Points for 'fixed' credits (threshold credits are scored from savings)	>= 0
capex	Upfront cost, Rs, extra over the code-minimum baseline	>= 0
subsidy	Subsidy received at year 0, Rs	<= capex
om_annual	Maintenance in year 1, Rs per year	>= 0
om_escalation	Yearly growth of maintenance (fraction)	-0.05 to 0.20
energy_saving_kwh	Electricity saved, kWh per year (efficiency measures)	>= 0
kwp	Solar size in kWp (renewable measures; energy = kWp x yield)	> 0 for solar
water_saving_kl	Water saved, kL per year	>= 0
degradation	Yearly loss of performance (fraction), e.g. 0.006 for panels	0 to 0.10
source	'library' or 'quote:<file>' (set automatically by the quote importer)	optional
description	Free text	optional

Example row: *HVAC_VRF, VRF air conditioning, group HVAC, efficiency, credit EE-C2, capex Rs 6,000,000, O&M Rs 200,000/yr, saves 72,000 kWh/yr, degradation 0.005.*

components.csv: parts that wear out

Column	Meaning	Rule
measure_id	Which measure the part belongs to	must exist in measures.csv
component	Name, e.g. 'Inverter'	required
cost	Price today, Rs (inflated to the year it is bought)	>= 0
life_years	Re-bought every this many years; unused life at the end is credited back	> 0

igbc_credits.csv: how points are earned

Column	Meaning	Rule
credit_id, category, name	Credit code (e.g. EE-C2), category and title	required
max_points	Cap on the points this credit can give	> 0
rule	**fixed** (measure gives its points_fixed) or **threshold_pct** (points from % saving)	one of the two
basis	For threshold credits: energy_eff, energy_re or water	needed for threshold_pct
thresholds	Steps as 'percent:points' joined by |, e.g. 3:1|6:3|9:5 means 3% saving gives 1 point, 6% gives 3...	needed for threshold_pct
mandatory	True if the rating is impossible without it	True / False

igbc_levels.csv: points needed per level

Column	Meaning
level	Name, e.g. Gold
min_points	Points needed to reach it
rank	Order from lowest (1) to highest

Vendor quote PDFs (optional)

A text-based PDF quotation. The importer drafts: vendor, item, capacity, quoted price, annual maintenance and one replaceable part with its cost and life. You add what a quote never states (energy saved, IGBC credit, the baseline cost to deduct) before it joins the library.

Checks the tool applies to inputs

- Every number is range-checked (for example the discount rate must be between 0 and 30%, lives must be above zero, a subsidy cannot exceed capex).
- Every link between files is checked: unknown credit IDs, parts pointing at missing measures, duplicate measure IDs, a target level that doesn't exist.
- Solar measures must have a kWp; threshold credits must have a basis and thresholds.
- All problems are reported together, in plain words with the file, row and field, so one fix-and-reload pass is enough.

What comes out

Output	Who it is for	What it contains
Answer on screen	Everyone	One sentence per page, number cards, charts with 'What this means' lines, Quick what-if sliders
Decision report (PDF, 6 pages)	Owner, professor, IGBC consultant, tender team	The answer and memo; the two routes and where the cost comes from; IGBC credit checklist with documents to submit; what to buy; risk check; assumptions and yearly cash flow
Results workbook (Excel)	Whoever checks the maths	Summary, checklist, shortlist, yearly cash flow, every simulated future, top 200 combinations, all inputs
Project file (.json)	You, next time	The whole setup in one file; open it from Home

Inside the repository

Folder / file	What is there
app.py	Starts the app: menu, sidebar controls, page routing
pages/	The five screens: 0_home, 1_setup, 2_recommendation, 3_trust, 4_report
core/	The engine: finance, cashflow, igbc, optimizer, risk, carbon, extract (quotes), report and exports (PDF/Excel), charts, ui, state, io, schemas
data/	The input files described above
sample_quotes/	Three illustrative vendor quotations
tests/	60 automated tests: hand-checked maths, IGBC scoring, optimiser, risk, exports, every screen
scripts/	Rebuild these documents, the sample quotes and the progress-log index
docs/	This document, the user guide and the project log

Honest limits

- **Illustrative data.** Prices, tariffs, IGBC point tables and the FAR bonus are placeholders until verified against official sources.
- **Savings add up simply.** In reality a better envelope reduces what efficient air-conditioning saves; a full energy simulation would refine this.
- **One tariff path and a fixed grid emission factor.** Real projects would use state tariff schedules and a falling emission factor.
- **The AI helpers need an API key** and have only been tested with their offline fallbacks so far.

Built for CM 617 (Topic 6, LCCA decision tool) by a five-member student team.